

Beyond Flattened Tokens: Structure-Preserving EEG Decoding with Reusable TriDim Blocks

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Abstract

Effective EEG decoding requires representations that preserve organization among channels, local waveform dynamics, and long-range temporal context. Existing EEG architectures often capture these structures using separate specialized modules or collapse them into a single token sequence, making it difficult to maintain their distinct roles and coordinate their interactions throughout the backbone. We propose TriDim, a reusable block that preserves the representation shape and keeps three EEG axes explicit: channel, sample position within each patch, and patch position across the recording. These axes correspond to spatial, short-term temporal, and long-term temporal information, respectively. Each TriDim block applies feed-forward transformations along individual axes and cross-axis attention to coordinate information exchange among them. By stacking TriDim blocks with a multi-level tri-axis readout, we construct TriDimEEG, a standalone EEG decoder. Under strict cross-subject evaluation on eight datasets spanning clinical diagnosis, sleep staging, motor imagery, and emotion recognition, TriDimEEG achieves the best overall performance among fifteen evaluated models, with a 4.3% relative improvement in average accuracy over the second-best model. Replacing Transformer blocks in three EEG foundation models with TriDim blocks yields an average relative improvement of 7.4% in downstream accuracy while reducing parameter counts by 17.0% to 47.3%. These results establish TriDim as an effective and reusable building block and TriDimEEG as a strong standalone EEG decoder.

Code and parameters of TriDimEEG are available at https://github.com/ncclab-sustech/TriDim_model.

Introduction

Electroencephalography (EEG) decoding supports diverse applications, such as clinical diagnosis, sleep monitoring, robotic control, and affective computing (Craik et al., 2019; Roy et al., 2019). EEG signals can be characterized by three complementary structural aspects (Figure 1): spatial organization across channels, waveform dynamics within local windows, and long-range temporal context across windows (Lawhern et al., 2018; Ding et al., 2023; Song et al., 2023; Wang et al., 2025a). Spatial organization captures how neural activity is distributed and coordinated across recording channels. Local waveform dynamics characterize oscillatory and transient patterns within short intervals. Long-range temporal context describes how these patterns evolve across extended periods. The three information types are also strongly interdepen-

dent (Wang et al., 2025a, 2026). For example, the same local waveform may indicate different neural states when expressed over different channel distributions, and the meaning of a spatial pattern may change according to the preceding and subsequent temporal context. Effective EEG decoding therefore requires a backbone that progressively refines each information type and repeatedly coordinates interactions among the three information types. Existing architectures, however, often prioritize only part of this structure and do not provide sufficiently deep and coordinated modeling of all three information types.

The main limitation lies in how current EEG architectures organize computation across the spatial and temporal dimensions of EEG. Compact convolutional networks use temporal kernels and spatial filters to capture lo-

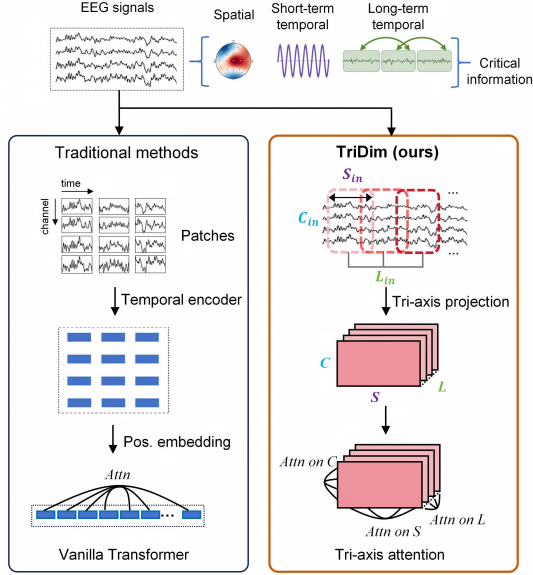


Figure 1. Comparison of conventional flattened-token EEG architecture and TriDim. TriDim preserves channel, within-patch temporal, and across-patch temporal structures as three explicit axes for tri-axis modeling.

cal dynamics and channel interactions (Lawhern et al., 2018; Ding et al., 2023; Miao et al., 2023), but predefined receptive fields constrain their ability to model long-range dependencies. Transformer-based models address this limitation in different ways. CNN-Transformer hybrids combine convolutional feature extraction with self-attention (Song et al., 2023; Wan et al., 2023; Ding et al., 2025). Many pretrained EEG encoders divide multichannel recordings into channel-wise temporal patches and arrange the resulting channel-time patches as a single token sequence for Transformer processing (Yang et al., 2023; Jiang et al., 2024; Wang et al., 2024). As illustrated in Figure 1, this conventional flattened-token pipeline mixes spatial and temporal structures within a single sequence-processing pathway. Unlike language, EEG is a continuous multichannel signal with temporal structure at multiple scales. A single token sequence assigns spatial organization and temporal dependencies to the same sequence-processing pathway. This design limits the depth and specificity with which the spatial, local temporal, and global temporal information and their interactions can be modeled. Criss-cross Transformers partly preserve EEG structure by applying self-attention along separate spatial and temporal directions (Wang et al., 2025a, 2026). However, these models still assign fixed computational roles to the three EEG axes. Channel and across-patch time are treated as attention axes, whereas ordered within-patch samples remain in the embedding

dimension. Consequently, within-patch representations are repeatedly refined by feed-forward networks without explicit relation modeling through self-attention. Channel and across-patch representations are repeatedly mixed through attention without axis-wise feed-forward transformation. Because self-attention and feed-forward networks provide complementary relational modeling and nonlinear representation refinement (Dong et al., 2021; Geva et al., 2020; Xu et al., 2024), this fixed assignment limits the deep processing of the three information types and the progressive refinement of their interactions throughout the backbone.

To enable deep processing of all three EEG information types and their interactions, we adopt a different design principle: channel, within-patch time, and across-patch time should remain explicit representation axes throughout the backbone. Based on this principle, we propose TriDim, a shape-preserving block for EEG modeling (Figure 2). TriDim constructs three axis-conditioned views. In each view, one axis serves as the embedding dimension, and self-attention is applied along the other two axes. The resulting views are combined through learnable weighted aggregation. Stacking TriDim blocks therefore enables spatial, short-term temporal, and long-term temporal information to undergo repeated feed-forward transformation, and interactions among the three information types are progressively updated through self-attention. A multi-level tri-axis readout further aggregates representations from different encoder depths. The resulting TriDimEEG model can serve as a task-specific EEG decoder, and the same TriDim block can also replace Transformer blocks in pretrained EEG encoders without redesigning the surrounding pipelines. This work makes three main contributions:

- We reinterpret temporally patched EEG representations using three explicit axes corresponding to channel, sample position within each patch, and patch position along the recording. This organization preserves spatial, short-term temporal, and long-term temporal information throughout the backbone.
- We develop TriDim, a reusable block that combines feed-forward transformations along individual axes with cross-axis attention. We further construct TriDimEEG by stacking TriDim blocks and introducing a multi-level tri-axis readout that aggregates representations across encoder depths.
- Strict cross-subject experiments on eight EEG benchmarks establish TriDimEEG as a strong standalone decoder. Replacing Transformer blocks in three EEG

foundation models further demonstrates the reusability and parameter efficiency of TriDim.

Related Work

Task-Specific EEG Decoders

Early EEG decoders primarily relied on convolutional networks to capture local waveform patterns and inter-channel relationships. EEGNet combines temporal convolution with depthwise spatial filtering in a compact architecture (Lawhern et al., 2018). Subsequent supervised models introduce multi-scale temporal kernels and specialized spatial processing for different EEG tasks (Ding et al., 2023; Miao et al., 2023; Ma et al., 2026). Attention-based architectures further extend this paradigm by modeling dependencies beyond local convolutional receptive fields. EEGConformer applies self-attention to features extracted by a convolutional temporal-spatial encoder (Song et al., 2023). EEGDeformer introduces hierarchical coarse-to-fine Transformer processing and dense information purification to aggregate representations across multiple levels (Ding et al., 2025). TeCh replaces conventional pairwise self-attention with Core Token Aggregation-Redistribution, which collects global context into a core token and redistributes it to the remaining tokens (Yu et al., 2026). Despite their architectural differences, these methods generally convert EEG signals into feature sequences before applying attention-based contextual modeling.

Pretrained EEG Representation Models

Large-scale pretraining has further established sequence modeling as a prominent paradigm for transferable EEG representation learning. BENDR (Kostas et al., 2021), BIOT (Yang et al., 2023), LaBraM (Jiang et al., 2024), and REVE (El Ouahidi et al., 2025) learn contextual representations from segmented EEG recordings through convolutional or Transformer-based encoders. Recent pretrained models additionally explore language-aligned representations (Cui et al., 2024), state-space sequence modeling (Wang et al., 2025b), frequency-enhanced encoders (Tegon et al., 2025), and sensor-aware tokenization (Xiao et al., 2025). These approaches have substantially improved transfer across datasets and downstream tasks. Nevertheless, despite differences in backbone architectures and pretraining objectives, most of these models still arrange local EEG segments into a one-dimensional token sequence, leaving spatial organization and across-segment temporal dependencies to be modeled within the same sequence-processing pathway.

Structure-Preserving EEG Architectures

Recent studies have begun to design attention mechanisms more explicitly aligned with the structural organization of EEG. CBraMod introduces a criss-cross Transformer that separately performs spatial attention across electrodes and temporal attention across windows (Wang et al., 2025a). BraSTORM adopts separate spatial and temporal branches to learn complementary representations through input-based spatiotemporal decomposition (Wang et al., 2026). CSBrain further introduces cross-scale spatiotemporal tokenization and structured sparse attention to model dependencies across temporal windows and anatomical regions (Zhou et al., 2025). These methods demonstrate the value of preserving structural distinctions in EEG rather than treating all signal components as positions in a homogeneous sequence. However, existing structure-aware architectures still assign fixed and asymmetric computational roles to the EEG axes, typically treating channel and across-patch time as attention axes and retaining within-patch samples as the embedding dimension. TriDim removes this fixed assignment by preserving channel, within-patch time, and across-patch time as three explicit axes, allowing each axis to undergo both axis-specific feed-forward transformation and cross-axis attention throughout the backbone.

Methods

TriDimEEG consists of a tri-axis projection, an encoder constructed from N stacked TriDim blocks, and a multi-level tri-axis readout.

Tri-axis Representation

We organize EEG representations along three axes: channel organization, short-term temporal structure, and long-term temporal context. Given an input batch $X_{\text{raw}} \in \mathbb{R}^{B \times C_{\text{in}} \times T_{\text{raw}}}$ with B EEG samples, C_{in} electrodes, and T_{raw} time points, we first apply InstanceTimeNorm independently to each channel over time. This operation removes the temporal mean and normalizes the amplitude scale of each channel without changing the tensor shape:

$$\hat{X} \in \mathbb{R}^{B \times C_{\text{in}} \times T_{\text{raw}}}.$$

We then unfold the time axis into L_{in} patches of length S_{in} with stride R , applying zero-padding at the end when necessary. This operation separates the original temporal dimension into the ordered samples within each patch and the ordered patch positions along the segment:

$$\hat{X} \in \mathbb{R}^{B \times C_{\text{in}} \times T_{\text{raw}}} \longrightarrow X^{\text{patch}} \in \mathbb{R}^{B \times C_{\text{in}} \times S_{\text{in}} \times L_{\text{in}}}.$$

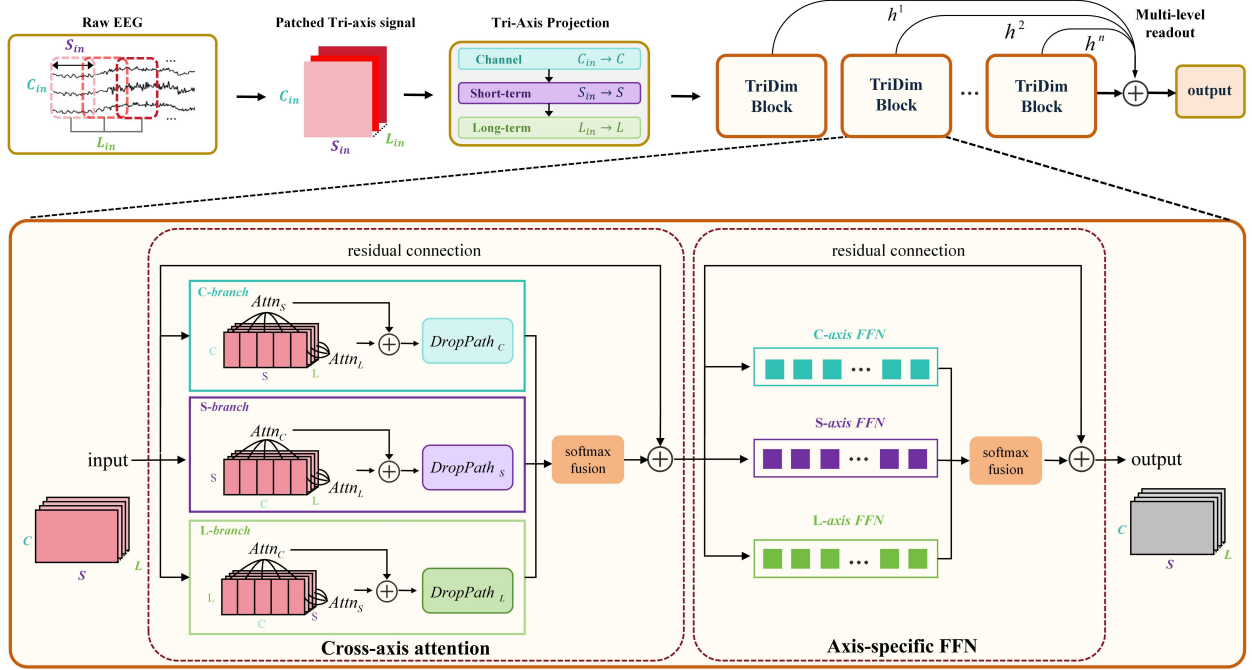


Figure 2. Overview of TriDimEEG, including tri-axis input projections, stacked TriDim blocks, and a multi-level tri-axis readout.

Three basis projections, one for each representation axis, map $(C_{\text{in}}, S_{\text{in}}, L_{\text{in}})$ to the latent dimensions (C, S, L) :

$$\begin{aligned} \mathbb{R}^{C_{\text{in}}} &\rightarrow \mathbb{R}^C \quad (1 \times 1 \text{ convolution}), \\ \mathbb{R}^{S_{\text{in}}} &\rightarrow \mathbb{R}^S \quad (\text{linear projection}), \\ \mathbb{R}^{L_{\text{in}}} &\rightarrow \mathbb{R}^L \quad (\text{shared convolution and pooling}). \end{aligned}$$

The channel projection applies a 1×1 convolution along the electrode axis C_{in} . The short-term projection applies a linear transformation along S_{in} . The long-term projection applies a shared one-dimensional convolution to the patch sequence at every channel and within-patch position, followed by GELU, dropout, and adaptive average pooling along L_{in} . The three projections produce

$$X^{\text{patch}} \in \mathbb{R}^{B \times C_{\text{in}} \times S_{\text{in}} \times L_{\text{in}}} \rightarrow X^0 \in \mathbb{R}^{B \times C \times S \times L}.$$

We use $\mathcal{A} = \{c, s, l\}$ to denote the latent channel, short-term, and long-term axes, whose sizes are C , S , and L , respectively. The TriDimEEG encoder operates on the resulting $[B, C, S, L]$ representation, which is preserved throughout all TriDim blocks.

TriDim Block

Let $X^\ell \in \mathbb{R}^{B \times C \times S \times L}$ denote the input to block $\ell + 1$, where $\ell = 0, \dots, N - 1$ and X^0 is defined above.

Each TriDim block contains two pre-normalized residual sublayers: a cross-axis attention sublayer followed by feed-forward transformations along individual axes. Both sublayers preserve the $[B, C, S, L]$ representation shape.

Cross-axis attention. For each axis $a \in \mathcal{A}$, we first apply Root Mean Square normalization (Zhang and Sennrich, 2019) along that axis:

$$\tilde{X}_a^\ell = \text{AxisRMSNorm}_a^{\text{attn}}(X^\ell), \quad a \in \mathcal{A}.$$

For a tensor Z , let $\text{Attn}_{p|q}(Z)$, where $p, q \in \mathcal{A}$ and $p \neq q$, denote multi-head self-attention along sequence axis p , with axis q serving as the embedding dimension. The remaining axis is treated as a collection of independent slices and folded into the batch dimension during attention computation.

For each embedding axis q , the two attention operations along the remaining axes share the same query, key, value, and output projection parameters. They are applied independently to the same normalized representation, and their outputs are averaged. The channel, short-term, and long-term views are computed as

$$\begin{aligned} V_c^\ell &= \frac{1}{2} \left[\text{Attn}_{s|c}(\tilde{X}_c^\ell) + \text{Attn}_{l|c}(\tilde{X}_c^\ell) \right], \\ V_s^\ell &= \frac{1}{2} \left[\text{Attn}_{c|s}(\tilde{X}_s^\ell) + \text{Attn}_{l|s}(\tilde{X}_s^\ell) \right], \\ V_l^\ell &= \frac{1}{2} \left[\text{Attn}_{c|l}(\tilde{X}_l^\ell) + \text{Attn}_{s|l}(\tilde{X}_l^\ell) \right]. \end{aligned}$$

Each view retains the $[B, C, S, L]$ shape and is named according to its embedding axis rather than the sequence axes over which attention is performed.

To robustly integrate the three complementary attention views, we regularize each view before fusion. Specifically, we apply DropPath independently to each view V_a^ℓ with dropout probability p_a^ℓ (Huang et al., 2016). This enables view-wise stochastic depth and discourages the model from relying excessively on any single EEG axis. The details are reported in Appendix G.

The regularized views are then combined using learnable fusion weights. Let

$$\alpha^\ell = \text{softmax}(\theta_{\text{attn}}^\ell) \in \mathbb{R}^3$$

denote the learnable view-fusion weights, and $\gamma_{\text{attn}}^\ell \in \mathbb{R}^C$ per-channel learnable LayerScale parameters (Touvron et al., 2021). The attention residual update is

$$X^{\ell+\frac{1}{2}} = X^\ell + \gamma_{\text{attn}}^\ell \odot \sum_{a \in \mathcal{A}} \alpha_a^\ell \text{DropPath}(V_a^\ell, p_a^\ell).$$

Axis-wise feed-forward sublayer. After cross-axis attention with multi-view fusion, an independent feed-forward network FFN_a is applied along each axis a . Each network contains two linear layers with an expansion ratio of 2, a GELU activation, and dropout. The intermediate representation is independently normalized along each target axis:

$$\begin{aligned} \bar{X}_a^{\ell+\frac{1}{2}} &= \text{AxisRMSNorm}_a^{\text{ffn}}(X^{\ell+\frac{1}{2}}), \\ U_a^\ell &= \text{FFN}_a(\bar{X}_a^{\ell+\frac{1}{2}}). \end{aligned}$$

The three axis-wise FFN outputs are first combined using learnable fusion weights:

$$\beta^\ell = \text{softmax}(\theta_{\text{ffn}}^\ell) \in \mathbb{R}^3,$$

which denotes the relative contributions of the three axes. DropPath is applied to the fused output. With per-channel learnable LayerScale parameters $\gamma_{\text{ffn}}^\ell \in \mathbb{R}^C$ and DropPath rate p_{ffn}^ℓ , the block output is

$$X^{\ell+1} = X^{\ell+\frac{1}{2}} + \gamma_{\text{ffn}}^\ell \odot \text{DropPath}\left(\sum_{a \in \mathcal{A}} \beta_a^\ell U_a^\ell, p_{\text{ffn}}^\ell\right).$$

Multi-level Tri-axis Readout

Representations from different encoder depths may contain complementary information. We therefore attach a tri-axis attention pooling head to every layer output X^ℓ , where $\ell \in \{1, \dots, N\}$.

For $X^\ell \in \mathbb{R}^{B \times C \times S \times L}$, the pooling head constructs one axis-specific sequence for each axis. The selected axis indexes the sequence positions, while the other two axes are flattened into the feature dimension:

$$\begin{aligned} Z_c^\ell &\in \mathbb{R}^{B \times C \times SL}, \\ Z_s^\ell &\in \mathbb{R}^{B \times S \times CL}, \\ Z_l^\ell &\in \mathbb{R}^{B \times L \times CS}, \end{aligned}$$

where $SL = S \cdot L$, $CL = C \cdot L$, and $CS = C \cdot S$.

An axis-specific linear projection W_a^ℓ maps each sequence to a shared embedding dimension E , after which a learnable positional embedding PE_a^ℓ is added:

$$H_a^\ell = Z_a^\ell W_a^\ell + \text{PE}_a^\ell \in \mathbb{R}^{B \times |a| \times E},$$

where $|a| \in \{C, S, L\}$ is the sequence length along axis a .

A lightweight attention pooling operator AttnPool_a^ℓ assigns normalized weights to the sequence positions and aggregates them into one E -dimensional vector. The three axis-specific vectors are combined to form the representation of layer ℓ :

$$h^\ell = \sum_{a \in \mathcal{A}} \omega_a^\ell \text{AttnPool}_a^\ell(H_a^\ell) \in \mathbb{R}^{B \times E},$$

where

$$\omega^\ell = \text{softmax}(\psi^\ell) \in \mathbb{R}^3$$

are learnable fusion weights.

Finally, learnable depth-fusion weights combine the representations from all N encoder layers:

$$\pi = \text{softmax}(\rho) \in \mathbb{R}^N, \quad z = \sum_{\ell=1}^N \pi_\ell h^\ell \in \mathbb{R}^{B \times E}.$$

The fused representation z is passed to a classifier head consisting of LayerNorm, dropout, and a linear classification layer.

Experimental Settings

Datasets and evaluation protocols. We evaluate TriDimEEG on eight EEG benchmarks covering four application scenarios: i) **Neurological disorder diagnosis:** AD65 (Miltiadous et al., 2023), (ii) **Sleep staging:** SleepEDF (Kemp et al., 2000; Goldberger et al., 2000), (iii) **Motor imagery:** BCI-IV-2A (Tangemann et al., 2012), SHU-MI (Ma et al., 2022), PhysioNet-MI (Schalk et al., 2004; Goldberger et al., 2000), and (iv) **Emotion recognition:** FACED (Chen et al., 2023), SEED (Zheng and Lu, 2015), and SEED-V (Liu et al., 2022). Table 1 summarizes the processed datasets.

Dataset	Subj.	Segments	Classes	Ch.	Samples
AD65	88	6,938	3	19	2,000
SleepEDF	153	414,961	5	2	6,000
BCI-2A	9	5,184	4	22	800
SHU-MI	25	7,571	2	32	800
PNet-MI	109	9,837	4	64	800
FACED	123	10,332	9	32	2,000
SEED	15	14,895	3	62	2,000
SEED-V	16	11,424	5	62	2,000

Table 1. Statistics of the processed EEG datasets. BCI-2A and PNet-MI stand for BCI-IV-2A and PhysioNet-MI, respectively. Subj. and Ch. denote subjects and channels.

We adopt a strict subject-independent evaluation protocol and divide subjects into training, validation, and test sets with an approximately 8:1:1 ratio. All recordings, sessions, and segments from the same subject are assigned to the same subset to prevent subject leakage. The subject-level partitioning procedure is repeated three times using different random seeds, yielding three predefined split manifests shared by all methods.

Preprocessing. All recordings are resampled to 200 Hz, band-pass filtered from 0.3 to 75 Hz, and notch filtered at the power-line frequency corresponding to each recording source. AD65, FACED, SEED, and SEED-V are divided into 10-s windows; BCI-IV-2A, SHU-MI, and PhysioNet-MI use 4-s windows; and SleepEDF uses standard 30-s sleep epochs following OmniEEG-bench (Lu et al., 2026).

Baselines. We compare TriDimEEG with four supervised EEG architectures: EEGNet (Lawhern et al., 2018), EEGConformer (Song et al., 2023), EEGDeformer (Ding et al., 2025), and TeCh (Yu et al., 2026). We also evaluate ten pretrained EEG models: BENDR (Kostas et al., 2021), BIOT (Yang et al., 2023), LaBraM (Jiang et al., 2024), CBraMod (Wang et al., 2025a), NeuroGPT (Cui et al., 2024), EEGMamba (Wang et al., 2025b), FEMBA (Tegon et al., 2025), NeuroLM (Jiang et al., 2025), BrainOmni (Xiao et al., 2025), and REVE (El Ouahidi et al., 2025). All methods use the same subject-level split manifests. For pretrained models, the complete pretrained encoder is fine-tuned jointly with a task-specific classification head on the training split. Necessary input adaptations follow the official implementations, with further details provided in Appendix E.

Results

We evaluate TriDim through four analyses: cross-subject benchmarking on eight EEG datasets, integration into three pretrained EEG encoders as replacing blocks, ab-

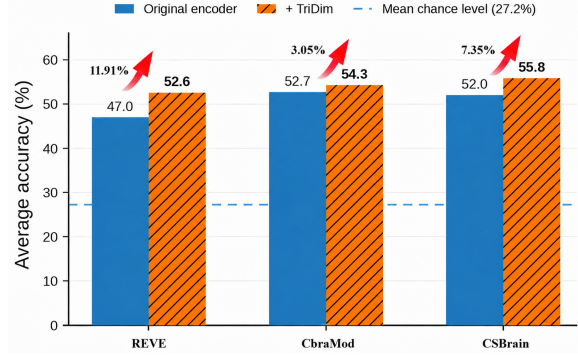


Figure 3. Replacing Transformer blocks with TriDim blocks under the same TUH-2K pretraining setting improves accuracy. Arrow annotations indicate relative improvements over the corresponding original encoder.

lation studies, and interpretation analysis. Results are reported as mean $\pm$ sample standard deviation over three subject-level splits, together with average rank across datasets. Macro-F1 results are provided in Appendix C.

Cross-Subject EEG Decoding Performance

Table 2 compares TriDimEEG with representative supervised and pretrained EEG models across clinical diagnosis, sleep staging, motor imagery, and emotion recognition tasks.

TriDimEEG achieves the highest average accuracy of 57.70% and the best average rank of 2.75, achieving a 4.3% relative improvement in average accuracy over the second-best model, EEGDeformer, and improving the average rank from 4.38 to 2.75. It ranks first on AD65, SleepEDF, BCI-IV-2A, and FACED, second on PhysioNet-MI, and third on SHU-MI. EEGDeformer remains strongest on SHU-MI and PhysioNet-MI, whereas REVE and LaBraM lead on SEED and SEED-V. However, these models are less consistent across the full benchmark. Although trained without large-scale EEG pretraining, TriDimEEG exceeds every evaluated pretrained encoder in both average accuracy and average rank.

Replacing Transformer Blocks in Pretrained EEG Encoders

To evaluate TriDim as a reusable architectural block, we replace the original Transformer-based blocks in REVE, CBraMod, and CSBrain (Zhou et al., 2025) with TriDim blocks. For each encoder, the original and TriDim-based variants are pretrained on the same 2,000 TUH samples using identical pretraining and downstream settings. Details are provided in Appendix F.

Replacing Transformer-based blocks with TriDim blocks yields relative improvements in average down-

Model	AD65	SleepEDF	BCI-2A	SHU-MI	PNet-MI	FACED	SEED	SEED-V	Avg. $\uparrow$	Rank $\downarrow$
EEGNet	59.6 $\pm$ 12.0	83.2 $\pm$ 2.2	32.8 $\pm$ 4.2	57.6 $\pm$ 6.4	57.5 $\pm$ 7.7	30.8 $\pm$ 1.2	45.0 $\pm$ 6.4	23.6 $\pm$ 2.8	48.76	8.38
EEGConformer	54.2 $\pm$ 10.7	84.0 $\pm$ 0.9	37.1 $\pm$ 2.4	56.8 $\pm$ 6.8	59.3 $\pm$ 6.6	38.7 $\pm$ 0.4	52.3 $\pm$ 7.6	25.1 $\pm$ 1.9	50.94	6.75
EEGDeformer	55.5 $\pm$ 12.5	83.1 $\pm$ 1.3	45.7 $\pm$ 2.6	68.7$\pm$13.4	62.5$\pm$8.6	45.9 $\pm$ 1.1	56.1 $\pm$ 6.4	25.2 $\pm$ 2.4	<u>55.32</u>	<u>4.38</u>
TeCh	60.1 $\pm$ 8.8	<u>85.5$\pm$1.5</u>	31.6 $\pm$ 0.7	<u>60.5$\pm$7.6</u>	56.2 $\pm$ 8.1	30.5 $\pm$ 0.8	51.5 $\pm$ 1.8	23.7 $\pm$ 2.4	49.94	7.75
BENDR	50.1 $\pm$ 4.2	65.5 $\pm$ 0.8	25.1 $\pm$ 0.2	51.8 $\pm$ 2.0	46.6 $\pm$ 3.9	13.5 $\pm$ 1.9	39.1 $\pm$ 8.2	22.8 $\pm$ 3.2	39.32	13.88
BIOT	53.7 $\pm$ 10.7	20.0 $\pm$ 0.0	25.0 $\pm$ 0.0	54.0 $\pm$ 2.0	29.1 $\pm$ 1.9	18.7 $\pm$ 1.1	51.2 $\pm$ 2.5	24.0 $\pm$ 3.5	34.46	12.88
LaBraM	59.6 $\pm$ 6.4	76.0 $\pm$ 1.6	32.9 $\pm$ 2.2	57.0 $\pm$ 1.3	52.3 $\pm$ 3.7	31.7 $\pm$ 1.0	59.3 $\pm$ 0.8	34.6$\pm$8.6	50.42	6.13
CBraMod	53.8 $\pm$ 5.8	74.2 $\pm$ 1.2	<u>46.3$\pm$6.8</u>	56.9 $\pm$ 0.8	56.6 $\pm$ 2.8	<u>51.8$\pm$1.4</u>	<u>59.7$\pm$2.4</u>	27.7 $\pm$ 5.0	53.38	5.13
NeuroGPT	46.7 $\pm$ 9.7	70.6 $\pm$ 0.9	35.3 $\pm$ 0.9	53.9 $\pm$ 1.5	50.3 $\pm$ 1.6	36.0 $\pm$ 1.1	56.3 $\pm$ 1.7	26.6 $\pm$ 4.3	46.97	9.25
EEGMamba	50.3 $\pm$ 7.3	68.5 $\pm$ 1.1	33.0 $\pm$ 0.7	52.9 $\pm$ 2.1	32.0 $\pm$ 1.2	25.2 $\pm$ 1.7	52.5 $\pm$ 3.1	24.2 $\pm$ 3.2	42.32	11.75
FEMBA	<u>61.2$\pm$14.8</u>	74.4 $\pm$ 1.8	42.2 $\pm$ 4.8	54.5 $\pm$ 0.5	53.7 $\pm$ 4.8	28.5 $\pm$ 0.5	55.8 $\pm$ 1.7	29.3 $\pm$ 5.1	49.95	6.88
NeuroLM	46.0 $\pm$ 5.9	73.4 $\pm$ 1.3	37.8 $\pm$ 0.4	51.2 $\pm$ 1.9	54.0 $\pm$ 3.3	49.0 $\pm$ 4.1	57.5 $\pm$ 3.2	30.7 $\pm$ 9.8	49.95	7.88
BrainOmni	45.4 $\pm$ 4.9	70.5 $\pm$ 2.1	38.3 $\pm$ 4.1	55.5 $\pm$ 0.5	36.7 $\pm$ 2.3	33.9 $\pm$ 2.6	49.4 $\pm$ 3.1	26.6 $\pm$ 2.1	44.52	10.38
REVE	51.3 $\pm$ 13.7	73.1 $\pm$ 2.3	40.2 $\pm$ 2.9	58.7 $\pm$ 2.5	56.7 $\pm$ 3.5	30.7 $\pm$ 2.3	64.5$\pm$6.5	<u>32.5$\pm$5.5</u>	50.96	5.88
TriDimEEG	63.8$\pm$6.1	85.6$\pm$2.2	56.6$\pm$2.7	60.2 $\pm$ 7.2	<u>60.4$\pm$8.0</u>	53.0$\pm$2.9	52.9 $\pm$ 2.2	29.0 $\pm$ 4.3	57.70	2.75

Table 2. Cross-subject classification accuracy (%) on eight EEG datasets. BCI-2A and PNet-MI stand for BCI-IV-2A and PhysioNet-MI, respectively. The best and second-best results in each column are highlighted in bold and underlined, respectively. Lower average rank indicates better overall performance. Results are reported as mean $\pm$ standard deviation. Averages and ranks are computed across all eight datasets.

stream accuracy of 11.91%, 3.05%, and 7.35% over the original REVE, CBraMod, and CSBrain encoders, while reducing parameters by 32.0%, 17.0%, and 47.3%, respectively. The mean of these three relative improvements is 7.4%, showing that TriDim is reusable across pretrained architectures while improving parameter efficiency.

Ablation Studies

We evaluate the contributions of two aspects of TriDimEEG: the three representation axes and the design choices within the TriDim backbone (Table 3). The first group removes or isolates the channel, short-term temporal, and long-term temporal axes. The second group evaluates the multi-level readout, cross-axis attention, axis-specific feed-forward networks, attention sharing, and parallel axis integration.

Removing the channel, short-term temporal, or long-term temporal axis decreases average accuracy by 2.3%, 0.5%, and 1.6%, respectively, relative to the Full model. The larger losses for C and L highlight the importance of using channel or long-term temporal information as the embedding dimensions, which were ignored in previous studies. Retaining only one-axis branch obtain 55.44%, 55.04%, and 54.46% for C , S , and L , respectively, all worse than the two-axis model or full TriDimEEG. Together, these results indicate three-axis representations provide complementary information.

For model design choices, using only the final layer lowers average accuracy by 0.7% relative to the Full

model. Removing cross-axis attention and sharing one FFN across axes cause relative reductions of 1.0% and 1.2%, respectively, supporting the advantage of combining cross-axis interaction and axis-specific transformation. Independent attention reaches a comparable performance of 57.69% but uses twice as many parameters in attention modules, showing that parameter sharing preserves accuracy efficiently. Sequential $C \rightarrow S \rightarrow L$ processing lowers average accuracy by 1.0% relative to the Full model, supporting the advantage of parallel integration over sequential processing.

Interpretation analysis

To examine whether the three TriDim axes capture physiologically meaningful and class-specific EEG representations, we conduct perturbation analyses along the channel, within-patch time, and across-patch time dimensions. These three perturbations characterize the spatial, local spectral, and temporal information used by TriDimEEG for motor imagery decoding, respectively.

We analyze class-conditional sensitivity on the patched representation $X^{\text{patch}} \in \mathbb{R}^{B \times C_{in} \times S_{in} \times L_{in}}$. Using 500-ms patches with a 50-ms stride, channel perturbation zeros one electrode across all patch positions, spectral perturbation removes one 2-Hz component within every local patch, and temporal perturbation zeros one complete patch position. We report the signed drop in true-class probability over nine leave-one-subject-out models, with 95% subject-bootstrap confidence intervals in Figure 4.

Variant	AD65	SleepEDF	BCI-2A	SHU-MI	PNet-MI	FACED	SEED	SEED-V	Avg.
w/o channel axis (C)	59.7 $\pm$ 1.6	85.3 $\pm$ 1.5	53.5 $\pm$ 5.7	62.7 $\pm$ 5.7	<u>62.7$\pm$9.0</u>	50.6 $\pm$ 1.3	51.8 $\pm$ 4.4	25.0 $\pm$ 1.3	56.40
w/o short-term axis (S)	62.1 $\pm$ 7.2	84.8 $\pm$ 2.9	54.8 $\pm$ 7.8	61.7 $\pm$ 6.7	63.0$\pm$8.3	52.0 $\pm$ 4.0	53.8 $\pm$ 2.3	27.0 $\pm$ 1.0	57.40
w/o long-term axis (L)	59.6 $\pm$ 4.7	85.3 $\pm$ 2.5	54.9 $\pm$ 7.3	61.8 $\pm$ 6.3	61.7 $\pm$ 8.7	51.3 $\pm$ 0.8	52.7 $\pm$ 1.2	27.0 $\pm$ 0.6	56.78
Channel axis only (C)	55.8 $\pm$ 11.1	84.8 $\pm$ 2.4	52.1 $\pm$ 9.3	61.2 $\pm$ 8.7	60.6 $\pm$ 6.1	51.0 $\pm$ 2.4	51.0 $\pm$ 5.2	27.0 $\pm$ 3.3	55.44
Short-term axis only (S)	50.5 $\pm$ 2.9	85.3 $\pm$ 1.8	54.2 $\pm$ 7.7	63.0 $\pm$ 11.2	59.6 $\pm$ 7.8	50.1 $\pm$ 2.3	52.0 $\pm$ 0.2	25.7 $\pm$ 1.4	55.04
Long-term axis only (L)	48.3 $\pm$ 1.8	84.7 $\pm$ 1.6	53.4 $\pm$ 6.7	59.0 $\pm$ 3.8	62.2 $\pm$ 7.5	50.2 $\pm$ 1.8	51.9 $\pm$ 0.6	26.0 $\pm$ 3.0	54.46
w/o multi-level readout	61.1 $\pm$ 4.7	86.0$\pm$2.4	55.7 $\pm$ 6.0	59.4 $\pm$ 5.3	62.4 $\pm$ 7.6	49.5 $\pm$ 2.3	<u>54.4$\pm$0.6</u>	<u>29.8$\pm$5.1</u>	57.30
w/o cross-axis attention	60.1 $\pm$ 9.4	84.2 $\pm$ 2.9	54.5 $\pm$ 4.3	62.9 $\pm$ 9.8	62.0 $\pm$ 8.6	52.0 $\pm$ 2.3	54.3 $\pm$ 3.1	27.0 $\pm$ 3.4	57.13
Shared FFN	<u>62.5$\pm$6.8</u>	84.8 $\pm$ 2.6	54.6 $\pm$ 5.9	61.8 $\pm$ 6.9	61.0 $\pm$ 6.4	50.9 $\pm$ 0.8	51.6 $\pm$ 1.8	28.8 $\pm$ 3.9	57.01
Independent attention	57.9 $\pm$ 7.2	85.4 $\pm$ 2.1	<u>55.8$\pm$5.1</u>	63.3 $\pm$ 8.0	61.7 $\pm$ 7.7	<u>52.4$\pm$1.6</u>	56.7$\pm$1.3	28.4 $\pm$ 6.1	<u>57.69</u>
Sequential $C \rightarrow S \rightarrow L$	58.3 $\pm$ 3.6	<u>85.9$\pm$2.0</u>	52.8 $\pm$ 7.8	64.0$\pm$5.8	60.5 $\pm$ 10.0	52.0 $\pm$ 0.9	52.4 $\pm$ 1.2	31.1$\pm$5.6	57.11
Full TriDimEEG	63.8$\pm$6.1	85.6 $\pm$ 2.2	56.6$\pm$2.7	60.2 $\pm$ 7.2	60.4 $\pm$ 8.0	53.0$\pm$2.9	52.9 $\pm$ 2.2	29.0 $\pm$ 4.3	57.70

Table 3. Ablation accuracy (% , mean $\pm$ standard deviation). BCI-2A and PNet-MI denote BCI-IV-2A and PhysioNet-MI. The best and second-best results in each column are highlighted in bold and underlined, respectively. Avg. averages all eight datasets.

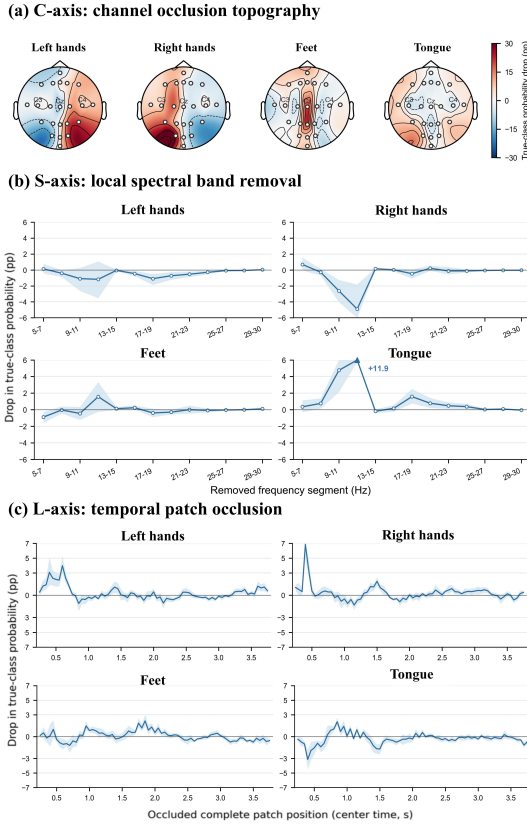


Figure 4. Class-conditional C/S/L-axis perturbations on BCI-IV-2A using channel, 2-Hz spectral, and 500-ms patch occlusion. Shading denotes 95% confidence intervals across nine held-out subjects.

Channel occlusion produces class-dependent scalp patterns (Figure 4(a)). The hand classes predominantly rely on contralateral centroparietal regions, whereas the feet class shows the strongest dependence near the central midline. **Spectral perturbation** shows its largest class-specific effects around 11–13 Hz (Figure 4(b)), overlapping the sensorimotor μ rhythm. The effect is strongly positive for tongue, moderately positive for feet, and negative for right hand, indicating that the same spectral component provides different evidence across classes. After aggregating the signed drops across classes within each subject, the overall response retains a clear 11–13 Hz peak (1.66 pp; 95% CI: 0.44–3.14 pp), while other frequencies remain close to zero. **Temporal patch occlusion** is strongest during the early trial period for the hand classes (Figure 4(c)). Feet exhibits a later peak around 1.8–2.0 s, whereas tongue shows an early negative response followed by weaker positive effects. Together, these results show that TriDimEEG can capture task-dependent spatial, local spectral, and temporal representations.

Conclusion

We introduced TriDim, a shape-preserving tri-axis block that jointly models channel organization, short-term waveform structure, and long-term temporal context through progressive axis-specific refinement and cross-axis interaction. Under strict cross-subject evaluation across eight EEG datasets, TriDimEEG achieved the highest average accuracy and best average rank among 15 models, with ablation and perturbation analyses confirming the complementary contributions of the three axes. As a drop-in replacement for Transformer blocks in three pretrained

EEG encoders, TriDim achieved a mean relative improvement of 7.4% in average downstream accuracy over the corresponding original encoders and reduced parameter counts. These results demonstrate that TriDim is an effective and reusable backbone block for task-specific and pretrained EEG models.

Acknowledgments

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Supplementary Material

A. Dataset and Preprocessing Details

Table 1 of the main text summarizes the eight processed datasets. All datasets are publicly available for research purposes and are cited in the main text. Raw recordings are preprocessed with a unified pipeline: resampling to 200 Hz, a 0.3–75 Hz band-pass filter, and a notch filter at the local power-line frequency. AD65, FACED, SEED, and SEED-V are segmented into 10-s windows; BCI-IV-2A, SHU-MI, and PhysioNet-MI use 4-s windows; SleepEDF uses standard 30-s sleep epochs. Per-sample, per-channel temporal normalization is performed by the InstanceTimeNorm layer at the model input, so no dataset-level statistics are computed or leaked across subjects.

For SEED-V we use the official 10-s benchmark segments; the label of each 10-s segment is inherited from its source 1-s annotation block. For SleepEDF we follow the supplied session-level protocol (153 sessions split 122/15/16) and retain at most 40 30-s epochs per session–sleep-stage pair in the training subset only (330,374 $\rightarrow$ 22,861 training epochs; the frozen manifests record the exact retained indices), which both bounds the training-set size and balances the five sleep stages (4,880/4,787/3,450/4,880/4,864 epochs for seed 5). Validation and test sessions are kept complete without truncation or downsampling (41,150 and 43,437 epochs), preserving the natural class distribution. The same retained-epoch manifests are used by every compared method on SleepEDF.

B. Evaluation Protocol

All methods are evaluated under a strict subject-independent protocol. Subjects are partitioned into training, validation, and test sets with an approximate 8:1:1 ratio, stratified by class where applicable; every recording, session, and segment of a subject is assigned to exactly one subset. Subject-level split manifests are generated before training, frozen as immutable JSON files, and shared by all compared methods, so every model is trained and tested on identical subject partitions. The frozen manifests for SEED-V and SleepEDF are shipped with the released code, together with their SHA-256 checksums; the remaining datasets use a seeded stratified subject splitter (train ratio 0.8, validation ratio 0.1) whose outputs are equally immutable for a given seed.

Each configuration is run under three independent subject-level splits generated by random seeds 5, 42, and 43. Within each run, the checkpoint with the highest validation accuracy is selected for final testing. We report the mean and sample standard deviation (ddof = 1) of test accuracy across the three splits. As aggregate measures across the eight datasets we report (i) average accuracy and (ii) average rank, where each model is ranked per dataset by mean accuracy (rank 1 = best) and the ranks are averaged; a lower average rank indicates better overall performance. Macro-F1 scores of the Full model and all ablation variants are reported in Supplementary Table 1; per-run records of every metric are included in the released result files.

Performance changes in the text are expressed relative to the stated reference: $100(A_{\text{new}} - A_{\text{ref}})/A_{\text{ref}}\%$, where A is the accuracy or macro-F1 for the comparison. For cross-dataset comparisons, A is the mean across the eight datasets. Ablation reductions use the Full model as the reference. Paired statistical differences and perturbation-induced probability changes retain their percentage-point units.

C. Macro-F1 Result Records

Supplementary Table 1 reports the macro-F1 score of the Full model and all eleven ablation variants on each of the eight datasets, as the mean $\pm$ sample standard deviation across the three subject-level splits (seeds 5/42/43); the underlying runs are exactly those behind Tables 2 and 3 of the main text. Per-split accuracy and macro-F1 records of every run, including the compared baselines, are included in the released result files. Average macro-F1 and average rank are computed across all eight datasets; a lower average rank indicates better overall cross-dataset ranking.

D. Leave-One-Subject-Out Evaluation

The 8:1:1 subject partition used in the main table is retained for consistency across all eight datasets, but on small-cohort datasets it yields test sets of one to three subjects. We therefore additionally evaluate the Full TriDimEEG model under leave-one-subject-out (LOSO) cross-validation on the four small-cohort datasets: BCI-IV-2A (9 subjects), SHU-MI (25 subjects), SEED (15 subjects), and SEED-V (16 subjects). BCI-IV-2A is the most affected case and is also the dataset

Supplementary Table 1. Macro-F1 (%) of Full TriDimEEG and all ablation variants on the eight datasets. Each cell reports the mean $\pm$ sample standard deviation (ddof = 1) across the three subject-level splits (seeds 5/42/43); the underlying runs are exactly those behind Tables 2 and 3 of the main text. The best and second-best results in each column are highlighted in bold and underlined, respectively.

Variant	AD65	SleepEDF	BCI-2A	SHU-MI	PNet-MI	FACED	SEED	SEED-V	Avg. $\uparrow$
w/o channel axis (C)	51.2 $\pm$ 8.05	65.9 $\pm$ 3.26	53.1 $\pm$ 5.89	62.2 $\pm$ 5.85	62.9 $\pm$ 8.93	51.0 $\pm$ 1.34	51.0 $\pm$ 5.20	18.0 $\pm$ 8.49	51.91
w/o short-term axis (S)	55.3 $\pm$ 13.8	65.9 $\pm$ 5.08	54.6 $\pm$ 7.58	61.6 $\pm$ 6.67	63.0$\pm$8.55	52.5 $\pm$ 3.97	52.8 $\pm$ 3.66	20.2 $\pm$ 10.3	53.24
w/o long-term axis (L)	54.8 $\pm$ 8.50	66.4 $\pm$ 4.93	54.5 $\pm$ 7.67	61.7 $\pm$ 6.26	61.7 $\pm$ 8.45	51.7 $\pm$ 0.810	51.9 $\pm$ 1.42	20.4 $\pm$ 10.5	52.89
Channel axis only (C)	44.8 $\pm$ 15.4	66.7 $\pm$ 3.99	51.7 $\pm$ 9.48	61.1 $\pm$ 8.70	60.6 $\pm$ 6.41	51.2 $\pm$ 2.60	49.8 $\pm$ 6.75	23.6 $\pm$ 2.38	51.19
Short-term axis only (S)	43.0 $\pm$ 4.61	66.1 $\pm$ 3.37	53.9 $\pm$ 7.85	62.6 $\pm$ 11.5	59.3 $\pm$ 8.12	50.4 $\pm$ 2.21	50.5 $\pm$ 2.50	17.9 $\pm$ 8.39	50.46
Long-term axis only (L)	40.8 $\pm$ 4.40	65.4 $\pm$ 3.39	53.0 $\pm$ 7.19	58.7 $\pm$ 4.04	62.5 $\pm$ 7.50	50.6 $\pm$ 1.63	51.3 $\pm$ 1.10	24.6 $\pm$ 2.63	50.86
w/o multi-level readout	54.9 $\pm$ 10.2	67.9$\pm$3.80	55.5 $\pm$ 5.97	59.3 $\pm$ 5.39	62.3 $\pm$ 7.71	49.8 $\pm$ 2.33	53.8 $\pm$ 0.950	28.0 $\pm$ 5.13	53.94
w/o cross-axis attention	<u>56.7$\pm$12.1</u>	65.1 $\pm$ 5.33	54.0 $\pm$ 4.25	62.6 $\pm$ 10.1	62.1 $\pm$ 8.67	52.3 $\pm$ 2.22	53.4 $\pm$ 3.33	26.9 $\pm$ 3.41	<u>54.14</u>
Shared FFN	56.2 $\pm$ 13.5	65.4 $\pm$ 4.50	53.7 $\pm$ 5.68	61.1 $\pm$ 7.53	61.2 $\pm$ 6.15	51.4 $\pm$ 0.740	51.2 $\pm$ 1.19	26.6 $\pm$ 3.54	53.35
Independent attention	49.2 $\pm$ 3.45	66.4 $\pm$ 4.02	55.5 $\pm$ 4.93	63.2 $\pm$ 7.89	61.7 $\pm$ 7.82	<u>52.7$\pm$1.74</u>	56.0$\pm$1.56	26.5 $\pm$ 7.21	53.90
Sequential $C \rightarrow S \rightarrow L$	51.6 $\pm$ 5.54	<u>67.5$\pm$3.01</u>	52.0 $\pm$ 8.16	63.9$\pm$5.87	60.6 $\pm$ 10.0	52.3 $\pm$ 0.950	51.6 $\pm$ 0.190	30.8$\pm$6.24	53.79
Full TriDimEEG	60.4$\pm$7.62	66.9 $\pm$ 3.17	56.1$\pm$3.11	59.5 $\pm$ 7.16	60.6 $\pm$ 7.83	53.3$\pm$3.06	51.2 $\pm$ 1.95	<u>28.2$\pm$4.54</u>	54.53

with the most established LOSO convention; we report its complete fold $\times$ seed grid below and summarize the remaining three datasets at the fold level.

Protocol. For fold $k \in \{1, \dots, 9\}$, subject A0 k is the test subject, subject A0 $((k \bmod 9) + 1)$ is the validation subject, and the remaining seven subjects form the training set. Every fold is trained under the same three seeds (5, 42, 43) and the identical per-dataset configuration as the main-table BCI-IV-2A runs (Supplementary Tables 5 and 6), with validation-based checkpoint selection. The fold manifests are frozen JSON files with disjoint train/validation/test trial indices, shipped with the released code. The LOSO study uses the fixed BCI-IV-2A trial asset of 5,088 trials (subject A04 contributes 480 trials; all other subjects 576); Table 1 of the main text reports the raw segment count (5,184 = 9×576), and the released trial asset excludes 96 trials of subject A04, yielding the fixed 5,088-trial set used by all BCI-IV-2A experiments. The perturbation analysis uses the same nine seed-42 LOSO subject partitions, but retrains a separate set of models with 500-ms patches and a 50-ms stride. These interpretation models are therefore distinct from the 200-ms-patch models reported in Supplementary Table 2.

Supplementary Table 2. LOSO cross-subject test accuracy (%) of Full TriDimEEG on BCI-IV-2A. Each cell is one trained model; mean $\pm$ std is computed across the three seeds per fold. The overall row reports the mean $\pm$ std across all 27 fold $\times$ seed runs.

Fold (test subj.)	seed 5	seed 42	seed 43	mean $\pm$ std
fold 1 (A01)	63.7	38.7	47.1	49.8 $\pm$ 12.7
fold 2 (A02)	45.8	39.6	41.8	42.4 $\pm$ 3.16
fold 3 (A03)	65.3	57.6	61.5	61.5 $\pm$ 3.82
fold 4 (A04)	45.4	41.7	43.3	43.5 $\pm$ 1.88
fold 5 (A05)	51.6	58.2	51.0	53.6 $\pm$ 3.97
fold 6 (A06)	48.3	45.5	47.7	47.2 $\pm$ 1.48
fold 7 (A07)	57.1	55.9	61.1	58.0 $\pm$ 2.72
fold 8 (A08)	56.8	50.9	52.1	53.2 $\pm$ 3.12
fold 9 (A09)	55.7	45.1	52.1	51.0 $\pm$ 5.38
Overall	per-seed fold means: 54.4 / 48.1 / 50.9			51.1 $\pm$ 7.49

Results. Supplementary Table 2 reports the complete fold $\times$ seed grid. TriDim obtains 51.1 $\pm$ 7.49% under LOSO, below its 8:1:1 result (56.6 $\pm$ 2.71%); this gap is expected because LOSO evaluates every subject, including the hardest ones, whereas the 8:1:1 protocol tests on a single drawn subject per split. Per-fold means range from 42.4% (A02) to

61.5% (A03), reflecting the well-documented subject difficulty spread of this benchmark. We report this LOSO study as a protocol-robustness check; the main-table protocol is unchanged for cross-dataset comparability.

LOSO on SHU-MI, SEED, and SEED-V. We additionally ran LOSO cross-validation of the Full model (seed 42, identical per-dataset configurations as the main table) on the three remaining small-cohort datasets, with every subject serving once as the test subject and no subject leakage between training and testing. Supplementary Table 3 summarizes the fold-level results. On all three datasets the LOSO accuracy is consistent with the 8:1:1 three-split results of the main table (60.2 ± 7.16 on SHU-MI, 52.9 ± 2.22 on SEED, and 29.0 ± 4.34 on SEED-V), indicating that the main conclusions are robust to the choice of subject-partition protocol.

Supplementary Table 3. LOSO cross-subject results of Full TriDimEEG (seed 42) on SHU-MI, SEED, and SEED-V. Accuracy and macro-F1 (%) are reported as mean $\pm$ std across folds (one test subject per fold). The 3-split column recalls the main-table accuracy for reference.

Dataset	LOSO acc.	LOSO macro-F1	3-split acc.
SHU-MI	60.8 ± 12.4	59.9 ± 12.8	60.2 ± 7.16
SEED	54.5 ± 7.93	51.8 ± 9.44	52.9 ± 2.22
SEED-V	29.3 ± 4.86	26.9 ± 6.87	29.0 ± 4.34

E. Baseline Compatibility and Adaptation Details

All baselines are evaluated under the same frozen subject-level manifests, the same validation-based checkpoint selection, and the same early-stopping protocol as TriDim. Input-shape adaptation follows each official implementation without altering the subject partition: recordings are resampled, filtered, and segmented by the unified pipeline (see the *Dataset and Preprocessing Details* section), and each baseline receives its inputs in the channel layout its official code defines. When a dataset montage cannot supply the channels or derivations a baseline requires, the corresponding inputs are zero-filled; no method-specific subject re-partitioning, re-segmentation, or label remapping is applied.

F. Pretrained-Encoder Integration: Protocol and Per-Split Results

Integration protocol. For each host encoder (REVE, CBraMod, CSBrain), the original backbone blocks are replaced with TriDim blocks and the modified encoder is *re-pretrained from scratch* with the host’s original pretraining objective (masked reconstruction). For a controlled comparison, both the original and the TriDim-based backbones are pretrained on the same 2,000 samples selected from the TUH corpus (TUH-2K), with identical pretraining schedules and otherwise unchanged components (input frontend, pretraining head, positional schemes). Downstream evaluation then loads the re-pretrained weights and fine-tunes *all* parameters end-to-end with a newly initialized linear classification head—no module is frozen, and shape-compatible host tensors are loaded with a non-strict state-dict mapping. Because this reduced TUH-2K pretraining set differs from the original released checkpoints used in Table 2 of the main text, the absolute numbers of this section are not directly comparable with the main table; the comparison of interest is original versus TriDim-based backbone within each encoder under identical settings.

Tensor interface. TriDim operates on a $[B, C, S, L]$ tensor; each host supplies its own grid mapping. For CBraMod, the backbone is natively four-dimensional $[B, C, P, d_{\text{model}}]$ with $d_{\text{model}} = 200$; the channel, patch, and d_{model} axes map directly onto the C , S , and L axes of TriDim, and the convolutional stem, FFT features, and ACPE positional encoding are retained. For CSBrain, the backbone tensor $[B, D, C, P]$ is mapped with the feature dimension D as the channel axis, the brain-region axis as S , and the patch axis as L . For REVE, the flattened token sequence is rearranged into a two-dimensional token grid $(b \ (c \ h) \ d \rightarrow b \ c \ h \ d)$, with the host’s Fourier 3D-coordinate and time-index positional embeddings retained on the tokens. In all three cases no information is discarded by the mapping; the token count and feature width of the host are preserved.

Result records. Supplementary Table 4 lists the per-dataset mean test accuracy of the original and TriDim-based encoders under the TUH-2K setting. Per-split values are included in the released result files.

Supplementary Table 4. Per-dataset mean test accuracy (%) of original and TriDim-based encoders under the TUH-2K pretraining setting, averaged over the three subject-level splits (seeds 5/42/43).

Dataset	REVE		CBraMod		CSBrain	
	orig.	+TriDim	orig.	+TriDim	orig.	+TriDim
AD65	51.4	55.4	53.0	59.1	51.4	55.9
SleepEDF	70.2	84.4	84.2	83.5	81.6	86.0
BCI-IV-2A	37.3	45.6	44.5	44.9	41.2	46.6
SHU-MI	59.1	61.2	57.6	60.3	59.1	63.5
PhysioNet-MI	57.5	57.8	57.6	59.4	56.5	59.0
FACED	43.2	48.5	65.5	57.8	65.6	63.7
SEED	40.3	38.6	37.8	41.6	39.0	42.7
SEED-V	16.7	29.1	21.4	27.8	21.9	29.3
Average	47.0	52.6	52.7	54.3	52.0	55.8

Under this controlled evaluation, TriDim-based re-pretraining yields relative improvements in average downstream accuracy of 11.91%, 3.05%, and 7.35% over the original REVE, CBraMod, and CSBrain encoders, respectively. For each encoder, the relative improvement is computed from its mean accuracy across the eight datasets; averaging the three encoder-level relative improvements gives 7.4%. This is not an average of per-dataset relative improvements. Parameter counts are reduced by 32.0%, 17.0%, and 47.3%—matching the values reported in the main text. Gains are consistent across task categories for REVE and CSBrain; the CBraMod variant trades small decreases on FACED and SleepEDF for gains on the remaining six datasets.

G. Complete Hyperparameter Configurations

Supplementary Tables 5 and 6 list the exact final hyperparameters used for the Full TriDimEEG model on every dataset, taken verbatim from the released per-dataset YAML configuration files. Dataset-specific values were selected exclusively according to validation accuracy during development; all ablation variants reuse the same per-dataset configuration with only the structural change described in the following section. The multi-level tri-axis readout is enabled in all main experiments.

H. Ablation Variant Definitions

All ablation variants share the Full model’s frontend (InstanceTimeNorm, patch unfolding, three basis projections), the multi-level tri-axis readout, the classifier head, and every training hyperparameter; only the indicated structural element differs. Parameter counts below are measured under the SEED configuration (Full = 3,251,600); exact counts vary slightly with per-dataset configurations.

- **w/o channel axis (C) / w/o short-term axis (S) / w/o long-term axis (L).** The indicated axis is removed from the block: its cross-axis attention branch, its axis-specific FFN, and the associated normalizations and DropPath are dropped, and the learned softmax view fusion is renormalized over the two surviving axis views.
- **Channel axis only (C) / short-term axis only (S) / long-term axis only (L).** Only the indicated axis is retained: the attention branch and the axis-specific FFN of the kept axis are unchanged, the corresponding branches of the other two axes are removed, and the view fusion degenerates to the single surviving view.
- **w/o multi-level readout.** The multi-level tri-axis readout is disabled (`use_multi_level_readout=False`); the per-level readout projections and the layer-fusion weights are removed, and the single tri-axis attention-pooling head is applied to the final block output only.
- **w/o cross-axis attention.** The entire cross-axis attention sublayer is removed (the three shared axis-attention modules, per-axis attention normalizations, attention view-fusion weights, attention LayerScale, and the per-branch DropPath), leaving a purely feed-forward tri-axis mixer: $x \leftarrow x + \gamma_{\text{ffn}} \odot \text{FFN}(x)$. −137,865 parameters.

Supplementary Table 5. Architecture hyperparameters of Full TriDimEEG per dataset. $C/S/L$: latent axis sizes (basis dims); D : patch embedding width; N : number of TriDim blocks; p_a : maximum DropPath rates of the attention views and the FFN branch ($p_c/p_s/p_l/p_{\text{ffn}}$). The last column states how the DropPath rate is applied across the N blocks: *constant* holds the rate at p_a in every block, while *linear* $0 \rightarrow p_a$ ramps it linearly from 0 to p_a across blocks.

Dataset	Channels	Patch len/stride	C	S	L	D	Heads	N
AD65	19	80/20	96	32	32	128	4	4
SleepEDF	2	80/20	32	16	16	128	4	4
BCI-IV-2A	22	40/10	96	32	32	128	4	3
SHU-MI	32	32/8	48	16	16	128	4	3
PhysioNet-MI	64	80/20	96	32	32	256	4	6
FACED	32	80/20	96	32	32	256	4	6
SEED	62	80/80	96	32	32	128	4	3
SEED-V	62	64/16	96	32	32	256	4	6

Dataset	Dropout	$p_c/p_s/p_l/p_{\text{ffn}}$	DropPath schedule
AD65	0.20	0.10/0.02/0.08/0.02	constant
SleepEDF	0.15	0.25/0.05/0.15/0.05	linear $0 \rightarrow p_a$
BCI-IV-2A	0.10	0.10/0.02/0.08/0.02	constant
SHU-MI	0.20	0.10/0.02/0.08/0.02	constant
PhysioNet-MI	0.30	0.25/0.05/0.15/0.05	linear $0 \rightarrow p_a$
FACED	0.15	0.10/0.02/0.08/0.02	constant
SEED	0.15	0.10/0.02/0.08/0.02	constant
SEED-V	0.20	0.10/0.02/0.08/0.02	constant

Supplementary Table 6. Training hyperparameters of Full TriDimEEG per dataset. All runs use AdamW, cosine learning-rate decay with a 5-epoch linear warm-up, gradient clipping at norm 1.0, early stopping on validation accuracy, and checkpoint selection by validation accuracy.

Dataset	Batch size	Learning rate	Weight decay	Max epochs	Patience
AD65	32	5×10^{-4}	0.02	100	20
SleepEDF	32	5×10^{-4}	0.05	50	50
BCI-IV-2A	16	5×10^{-4}	0.01	100	20
SHU-MI	64	1×10^{-4}	0.02	100	15
PhysioNet-MI	32	5×10^{-4}	0.02	100	20
FACED	32	5×10^{-4}	0.00	100	20
SEED	32	5×10^{-4}	0.01	100	20
SEED-V	32	5×10^{-4}	0.05	50	50

- **Shared FFN.** The three axis-specific FFNs are replaced by one shared two-layer MLP (same expansion ratio 2). Each axis view is transposed so the target axis becomes the last dimension, right-zero-padded up to the channel-axis width, passed through the shared MLP, and sliced back; padding only ever widens the S/L views, so no view is truncated. $-25,152$ parameters.
- **Independent attention.** The three parameter-shared attention modules are replaced by six independent modules, one per (embedding axis, sequence axis) pair; every other component is unchanged. $+137,088$ parameters.
- **Sequential $C \rightarrow S \rightarrow L$.** The three axis views are applied sequentially in the fixed order $C \rightarrow S \rightarrow L$ instead of in parallel with learned softmax fusion; each sub-update reads the tensor updated by the previous one. The view-fusion weights are removed (-18 scalar parameters). During training, one of the three attention sub-updates

is randomly skipped per batch with the configured probability and the surviving two are scaled by 1.5, keeping the expected update identical to the parallel design.

I. Statistical Analysis

Variation across splits. Every entry in Tables 2 and 3 of the main text is the mean $\pm$ sample standard deviation over the three independent subject-level splits (seeds 5/42/43), rather than a single run; the corresponding macro-F1 scores are listed in Supplementary Table 1, and the underlying per-split values are included in the released result files.

Cross-model omnibus test. A Friedman test across the 15 compared models and the eight datasets (mean accuracy per dataset as the observation unit) rejects the null hypothesis of equal model performance ($\chi^2(14) = 55.84$, $p < 0.001$), confirming that the per-dataset ranking differences are systematic rather than random.

Cross-dataset pairwise tests. We compare TriDim with the strongest baselines using a two-sided Wilcoxon signed-rank test paired by dataset ($n = 8$, exact). TriDim wins 5/8 datasets against the second-best model EEGDeformer ($W = 11$, $p = 0.383$), 6/8 against REVE ($W = 6$, $p = 0.109$), and 7/8 against CBraMod ($W = 5$, $p = 0.078$). With only eight paired datasets the test has limited power (the smallest attainable two-sided p is 0.0078, and only for an 8/8 win), so we interpret the aggregate performance improvements (4.3% relative improvement in average accuracy over EEGDeformer, together with the better average rank) as the primary evidence, and report the test outcomes for completeness rather than as the basis of our claims.

Split-level tests for ablations. For the ablation study we exploit the fact that every variant is evaluated on the *same* frozen split manifests as the Full model: we pair per-split test accuracies by (dataset, split), giving $n = 8 \times 3 = 24$ paired observations per variant, and apply a two-sided Wilcoxon signed-rank test (normal approximation with continuity correction). Supplementary Table 7 reports the mean paired difference (Full – variant) and the test outcome. Removing any single axis does not significantly degrade performance ($p = 0.089$ for C , a marginal trend; $p = 0.70$ for S ; $p = 0.62$ for L), whereas restricting the model to a single axis does: the S -only and L -only variants are significantly worse than Full ($p = 0.012$ and $p = 0.015$), with C -only a marginal trend ($p = 0.10$). The structural variants (multi-level readout, cross-axis attention, shared FFN, sequential processing) each reduce average accuracy by approximately 0.7–1.2% relative to Full without reaching significance, and independent (unshared) attention performs essentially on par with Full, indicating that parameter sharing is performance-neutral. We therefore describe the tri-axis design as improving cross-task robustness—the Full model is the most consistent variant across datasets rather than the per-dataset winner—and avoid claiming per-variant significant differences where the paired test does not support them.

Supplementary Table 7. Split-level paired Wilcoxon signed-rank tests, Full TriDimEEG versus each ablation variant, paired by (dataset, split) with $n = 24$. Mean Δ is the mean paired accuracy difference (Full – variant) in percentage points; positive values favor Full. Mean Δ is computed from unrounded per-split values and may therefore differ by up to 0.01 pp from the difference of the rounded table entries.

Variant	Mean Δ (pp)	Wilcoxon p
w/o channel axis (C)	+1.30	0.089
w/o short-term axis (S)	+0.31	0.700
w/o long-term axis (L)	+0.92	0.617
Channel axis only (C)	+2.26	0.100
Short-term axis only (S)	+2.66	0.012
Long-term axis only (L)	+3.24	0.015
w/o multi-level readout	+0.40	0.786
w/o cross-axis attention	+0.57	0.637
Shared FFN	+0.69	0.539
Independent attention	+0.00	0.353
Sequential $C \rightarrow S \rightarrow L$	+0.59	0.867

J. Sensitivity Analysis Methodology

Perturbation analysis. The axis-aligned perturbation study shown in Figure 4 uses nine seed-42 LOSO models trained specifically for the interpretation analysis with 500-ms patches and a 50-ms stride. These models use the same LOSO subject partitions as the protocol analysis above, but differ from the 200-ms-patch BCI-IV-2A models used in the main benchmark and Supplementary Table 2. Each subject is scored by a model that never saw that subject during training or checkpoint selection. Perturbations are applied after patchification to the patched representation rather than directly to the raw EEG signal. Three perturbation families are evaluated class-conditionally: (i) *electrode occlusion*, zeroing one of the 22 channels across all patch positions; (ii) *frequency-band occlusion*, removing one 2-Hz spectral component within each 500-ms local patch; and (iii) *temporal patch occlusion*, zeroing one complete 500-ms patch while leaving the overlapping neighboring patches unchanged. With a 50-ms stride over each 4-s trial, the temporal analysis contains 71 patch positions. We record the signed drop in true-class probability (in percentage points) relative to the unperturbed input. For each condition, the subject-level mean drops are pooled across the nine held-out subjects, and 95% confidence intervals are obtained from a 10,000-draw bootstrap over subjects ($n = 9$). The complete per-condition summary tables are included in the released result files.

Alignment with the main configuration. The interpretation analysis deliberately uses a 500-ms patch length with a 50-ms stride, whereas the main BCI-IV-2A benchmark uses 200-ms patches with the same 50-ms stride. The 500-ms configuration provides the patch representation used for all three perturbation analyses in Figure 4 and yields 71 temporal occlusion positions over each 4-s trial.

Shuffled-label controls. As a negative control, we retrain the same nine LOSO models with the *training* labels randomly permuted (fixed permutation, seed 42; validation and test labels untouched) under the identical configuration. These models perform at chance on the held-out subjects (mean test accuracy 26.1% over the nine folds, versus 25% chance), confirming that no task-relevant structure is learned. Their perturbation profile collapses: the mean absolute true-class-probability drop shrinks from 5.71 to 0.71 pp for electrode occlusion, from 0.41 to 0.14 pp for frequency-band occlusion, and from 0.56 to 0.21 pp for patch occlusion, and the largest single-condition effect falls from 22.5 pp to at most 2.5 pp. The class-conditional spatial, spectral, and temporal structure reported in the main text is therefore absent under shuffled labels and cannot be attributed to the perturbation procedure itself.

K. Computing Infrastructure and Software

Experiments were conducted on two Linux GPU clusters. The main-table results were produced primarily on an internal cluster equipped with Tesla V100 (32 GB) GPUs (seven of the eight datasets), using Python 3.10, PyTorch 2.4.1 (CUDA 12.1, cuDNN 9.1), NumPy 2.1.2, and Zarr 2.18.3. The SleepEDF main-table runs, all block-ablation runs, and the release validation were executed on Volcano Engine ML Platform nodes equipped with 4×NVIDIA H20 (96 GB) GPUs per node, with one training process pinned to each GPU, using Python 3.11, PyTorch 2.8.0 (CUDA 12.8), and Zarr 2.18.7. Per-run logs record the exact environment string (`torch/cuda/zarr` versions), the resolved data root, the split manifest SHA-256, and a configuration check line (`multi_level_readout=True`, selection metric) for every training run.

L. Independent Reproduction Check

Before submission, the released code and configurations were re-executed from scratch on a different hardware and software stack than the one used for the primary results (H20 nodes with PyTorch 2.8.0/CUDA 12.8, versus V100 with PyTorch 2.4.1/CUDA 12.1). Three checks were performed. First, the released Full model builds to exactly 3,251,600 parameters under the SEED configuration, identical to the original training code, and the Full model together with all structurally distinct variant classes defined in the *Ablation Variant Definitions* section pass a forward/backward smoke test with gradients flowing to every parameter. Second, an end-to-end re-run of BCIC2A and SHU (three seeds each, full training protocol) reproduces the main-table mean accuracies within the original three-seed standard-deviation bands: 54.8 versus 56.6 ± 2.71 on BCIC2A, and 61.0 versus 60.2 ± 7.16 on SHU. Third, every reported number is machine-audited against the per-run result records (`release_audit/traceability_audit.py`, deviations below 10^{-6}). Finally, the same cleaned release was re-executed on the *original* V100/PyTorch 2.4.1 stack for BCIC2A (three seeds, full protocol): the per-split test accuracies (53.5/58.3/58.0, mean 56.6) match the original result records to the reported precision, confirming that the release reproduces the primary results under the original environment and that

the cross-stack deviations described below are environmental.

Training is not run in deterministic mode. Across heterogeneous GPU architectures and library versions, per-seed accuracies on small datasets can vary across runs, because validation-based early stopping amplifies kernel-level nondeterminism into different selected checkpoints; the three-seed mean and its spread are stable. This is why all results in this paper are reported as mean $\pm$ sample standard deviation over three seeds rather than per-seed point values.

M. Code Availability

The code repository for TriDimEEG is https://github.com/ncclab-sustech/TriDim_model. The release artifacts described in this supplement comprise the source code, per-dataset YAML configurations, frozen split manifests with SHA-256 checksums, per-seed result records and training logs, and a release-validation script (`scripts/validate_release.py`). Repository contents and licensing terms should be consulted before reuse. The validation script checks the dataset registry, configuration consistency, immutable manifests, and result-file completeness of the release.

N. Additional Architectural Controls

Beyond the ablation variants reported in the main text, we evaluate two additional architectural controls that isolate the contribution of the tri-axis factorization itself. Both controls reuse the frozen subject-level split manifests, the per-dataset configurations of Supplementary Tables 5 and 6, and the identical training protocol (seeds 5/42/43, validation-based checkpoint selection).

Criss-cross control. This control fixes the short-term axis S as the embedding dimension and applies only two attention operators, $\text{Attn}_{c|s}$ over the channel axis and $\text{Attn}_{l|s}$ over the long-term axis, both with S as the embedding axis. The two operators share parameters, and their outputs are combined by a fixed 0.5 average in a direct residual connection—there is no axis-role rotation and no learned attention view fusion. The FFN path (three axis-specific MLPs with learned softmax fusion) is identical to the Full model. The control uses 3,121,800 parameters versus 3,246,609 for the Full model under the same configuration (-3.84%), so the comparison is not confounded by capacity.

Supplementary Table 8 reports the per-dataset results. The criss-cross control falls below the Full model in accuracy on five of the eight datasets; the exceptions are SHU-MI, SleepEDF, and PhysioNet-MI, with relative improvements of approximately 7.6%, 0.3%, and 1.0% over Full, respectively. Across all eight datasets, it reduces average accuracy by approximately 3.1% and average macro-F1 by 4.0% relative to Full. A split-level paired Wilcoxon signed-rank test ($n = 8 \times 3 = 24$) does not reach significance (accuracy $p = 0.143$; macro-F1 $p = 0.107$), so we do not claim a significant per-pair degradation; the relevant observation is the consistency of the gap across datasets rather than its size on any single one.

Supplementary Table 8. Criss-cross control versus Full TriDimEEG: test accuracy and macro-F1 (%), reported as mean $\pm$ sample standard deviation across the three subject-level splits (seeds 5/42/43).

Dataset	Accuracy		Macro-F1	
	Full	criss-cross	Full	criss-cross
AD65	63.8 $\pm$ 6.10	52.8 $\pm$ 4.83	60.4 $\pm$ 7.62	47.4 $\pm$ 4.88
SleepEDF	85.6 $\pm$ 2.19	85.9 $\pm$ 2.51	66.9 $\pm$ 3.17	66.7 $\pm$ 4.82
BCI-IV-2A	56.6 $\pm$ 2.71	54.5 $\pm$ 8.34	56.1 $\pm$ 3.11	54.2 $\pm$ 8.58
SHU-MI	60.2 $\pm$ 7.16	64.8 $\pm$ 7.40	59.5 $\pm$ 7.16	64.3 $\pm$ 7.17
PhysioNet-MI	60.4 $\pm$ 8.00	61.0 $\pm$ 7.67	60.6 $\pm$ 7.83	61.2 $\pm$ 7.40
FACED	53.0 $\pm$ 2.87	52.0 $\pm$ 2.60	53.3 $\pm$ 3.06	52.4 $\pm$ 2.51
SEED	52.9 $\pm$ 2.22	50.6 $\pm$ 0.660	51.2 $\pm$ 1.95	47.8 $\pm$ 2.14
SEED-V	29.0 $\pm$ 4.34	25.5 $\pm$ 2.53	28.2 $\pm$ 4.54	24.2 $\pm$ 3.92
Average	57.7	55.9	54.5	52.3

Flattened Transformer control. This control replaces the tri-axis encoder with a standard Transformer operating on a flattened token sequence. Given the projected representation $\mathbf{X} \in \mathbb{R}^{B \times C \times S \times L}$, the short-term and long-term axes are flattened into a single sequence dimension, yielding $\mathbf{X}_{\text{flat}} \in \mathbb{R}^{B \times (S \cdot L) \times C}$. Thus, each (s, l) position forms one token, with the C -dimensional channel representation serving as its embedding. The flattened sequence is processed by the same number of standard pre-norm Transformer layers as TriDimEEG, each consisting of multi-head self-attention followed by a two-layer FFN with expansion ratio 2. The number of attention heads, dropout rate, and LayerScale initialization are matched to the Full model.

For compatibility with the unchanged multi-level tri-axis readout, the output of each Transformer layer is reshaped back to $[B, C, S, L]$ before readout. All components outside the encoder—including the frontend, tri-axis projections, multi-level tri-axis readout, and classifier head—are kept identical to the Full model. As shown in Supplementary Table 9, replacing the structure-preserving TriDim encoder with this flattened-token Transformer reduces average accuracy by approximately 1.2% and average macro-F1 by 3.1% relative to the Full model.

Supplementary Table 9. Flattened Transformer control versus Full TriDimEEG: test accuracy and macro-F1 (%), reported as mean $\pm$ sample standard deviation across the three subject-level splits (seeds 5/42/43).

Dataset	Accuracy		Macro-F1	
	Full	Flattened	Full	Flattened
AD65	63.8 $\pm$ 6.10	59.6 $\pm$ 12.4	60.4 $\pm$ 7.62	51.1 $\pm$ 16.7
SleepEDF	85.6 $\pm$ 2.19	84.8 $\pm$ 1.92	66.9 $\pm$ 3.17	66.0 $\pm$ 3.93
BCI-IV-2A	56.6 $\pm$ 2.71	49.8 $\pm$ 5.82	56.1 $\pm$ 3.11	48.9 $\pm$ 5.64
SHU-MI	60.2 $\pm$ 7.16	64.8 $\pm$ 9.13	59.5 $\pm$ 7.16	63.6 $\pm$ 10.2
PhysioNet-MI	60.4 $\pm$ 8.00	61.1 $\pm$ 6.55	60.6 $\pm$ 7.83	60.9 $\pm$ 6.28
FACED	53.0 $\pm$ 2.87	53.2 $\pm$ 2.99	53.3 $\pm$ 3.06	53.5 $\pm$ 2.98
SEED	52.9 $\pm$ 2.22	55.5 $\pm$ 2.54	51.2 $\pm$ 1.95	55.1 $\pm$ 3.12
SEED-V	29.0 $\pm$ 4.34	27.0 $\pm$ 6.51	28.2 $\pm$ 4.54	23.0 $\pm$ 9.68
Average	57.7	57.0	54.5	52.8

An ordered evidence chain. Taken together with the structural ablations of the main text, the architecture controls form an ordered evidence chain: Full TriDimEEG (57.7) > sequential $C \rightarrow S \rightarrow L$ (57.1) > Flattened Transformer (57.0) > criss-cross (55.9) in average accuracy. Removing the axis-role rotation while keeping two attention operators (criss-cross) costs more than removing the tri-axis factorization while keeping full self-attention (Flattened Transformer), and both fall below the Full tri-axis block, supporting the view that the structured three-axis decomposition—not merely the presence of attention—underlies the cross-task robustness of the Full model.